

WALMART SALES PERFORMANCE DASHBOARD

Using Microsoft Power BI

Project Report

Submitted By

Name: Athira S

Tool Used: Microsoft Power BI Desktop

1. Introduction

For this project, I built a sales performance dashboard in Power BI using Walmart's weekly sales data. I picked this dataset mainly because it has a good mix of numerical and categorical fields, store-level sales, dates, fuel prices, CPI, unemployment, and a holiday flag, so it gave me enough room to try out different chart types instead of just building a single summary table.

The idea behind the dashboard was to turn a fairly large, unfiltered spreadsheet into something a manager could actually glance at and understand in a minute. Once it was built, the dashboard let me filter the data by year, store, and holiday status, and see how those filters changed the sales numbers in real time.

2. Objectives

Going in, I set out with a few clear goals for what I wanted the dashboard to do:

- Analyse Walmart's weekly sales across all 45 stores.
- Build something interactive rather than a static report.
- Compare sales on holiday weeks versus regular weeks.
- Track how sales moved over time.
- Check whether fuel prices had any real effect on weekly sales.

3. Dataset Description

The dataset I used contains weekly sales records collected across Walmart's stores. The fields I actually worked with were:

- Store — store identifier (1 to 45)
- Date — the week the sales were recorded
- Weekly Sales — total sales for that store in that week
- Fuel Price — average fuel price for that week
- CPI — Consumer Price Index

- Unemployment — unemployment rate
- Holiday Flag — whether that week included a holiday

4. Data Preparation

Before I got into building any visuals, I spent some time cleaning up the dataset. The Date column, for instance, wasn't recognized as a proper date type when I first imported it, so I had to fix that in Power Query before anything downstream would work correctly. Here's roughly what I did:

- Imported the Excel file into Power BI Desktop.
- Corrected the date format so Power BI would read it as a proper date field.
- Checked that Weekly Sales, Fuel Price, CPI, and Unemployment were all recognized as numeric fields.
- Set up the DAX measures needed to drive the KPI cards.

5. DAX Measures Used

I created four core measures to power the KPI cards at the top of the dashboard:

- Total Sales
- Average Sales
- Store Count
- Maximum Sales

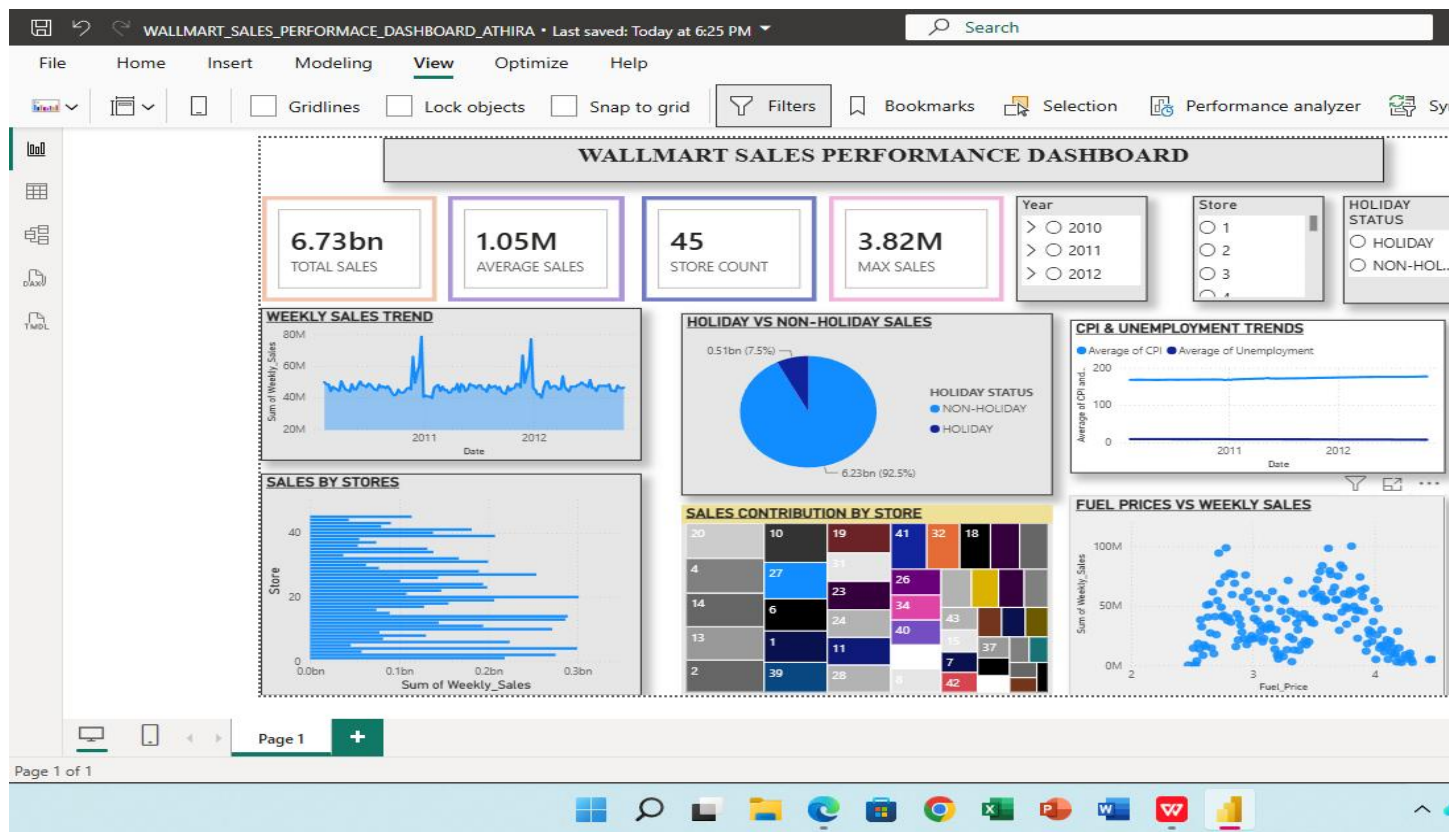
These were simple SUM, AVERAGE, DISTINCTCOUNT, and MAX measures built off the Weekly Sales column, but they're what make the top row of the dashboard readable at a glance.

6. Dashboard Components

The finished dashboard is made up of the following visuals.

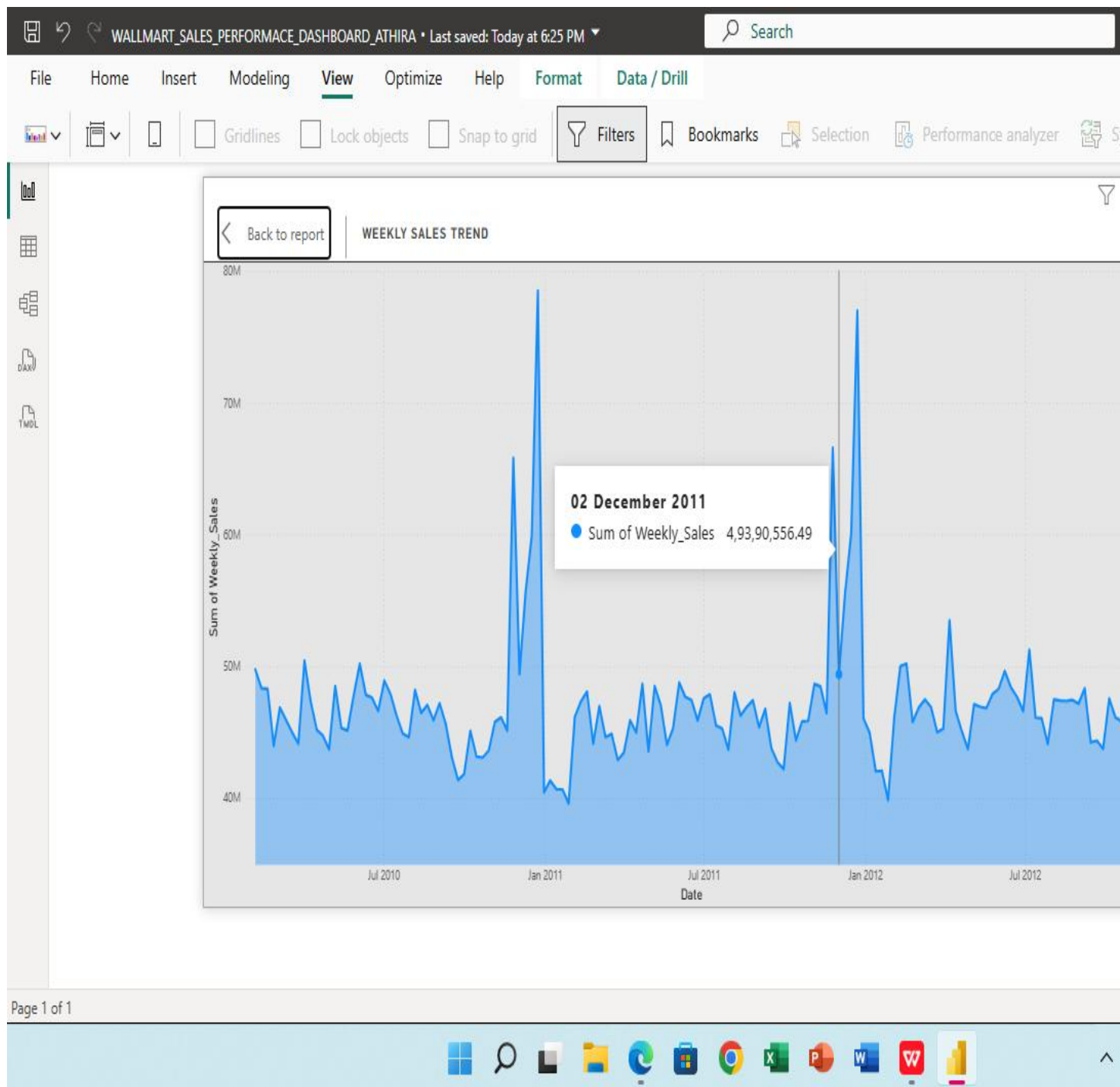
KPI Cards

Sitting at the top of the page, four cards show Total Sales, Average Sales, Number of Stores, and Maximum Weekly Sales. This was the first thing I built, since it gave me a quick sanity check that my measures were working before I moved on to the more complex charts



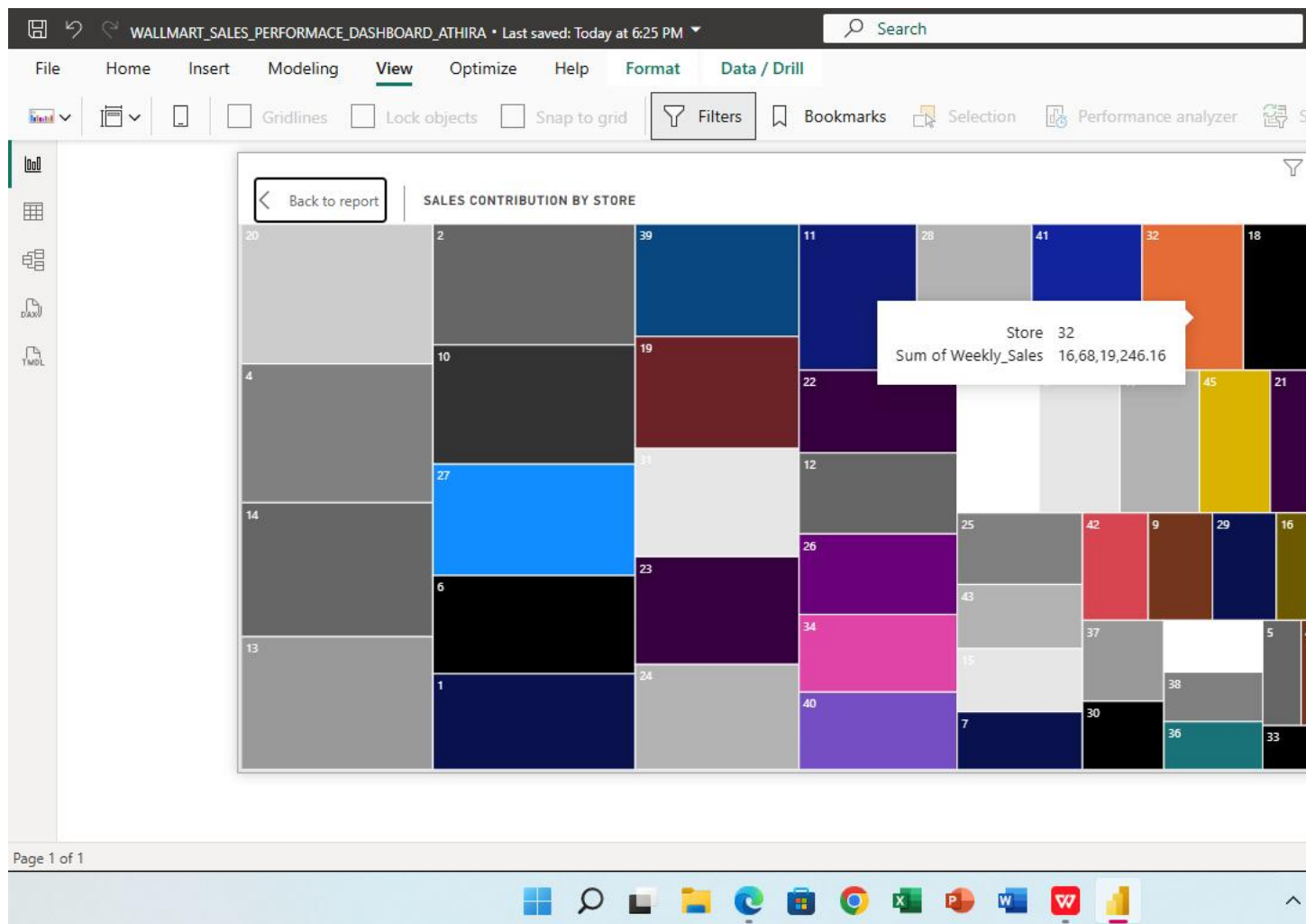
Weekly Sales Trend

A line chart tracks weekly sales across the full time period in the dataset. It made the seasonal spikes around the holiday season obvious almost immediately — sales climb noticeably in the run-up to major holidays and drop back down right after.



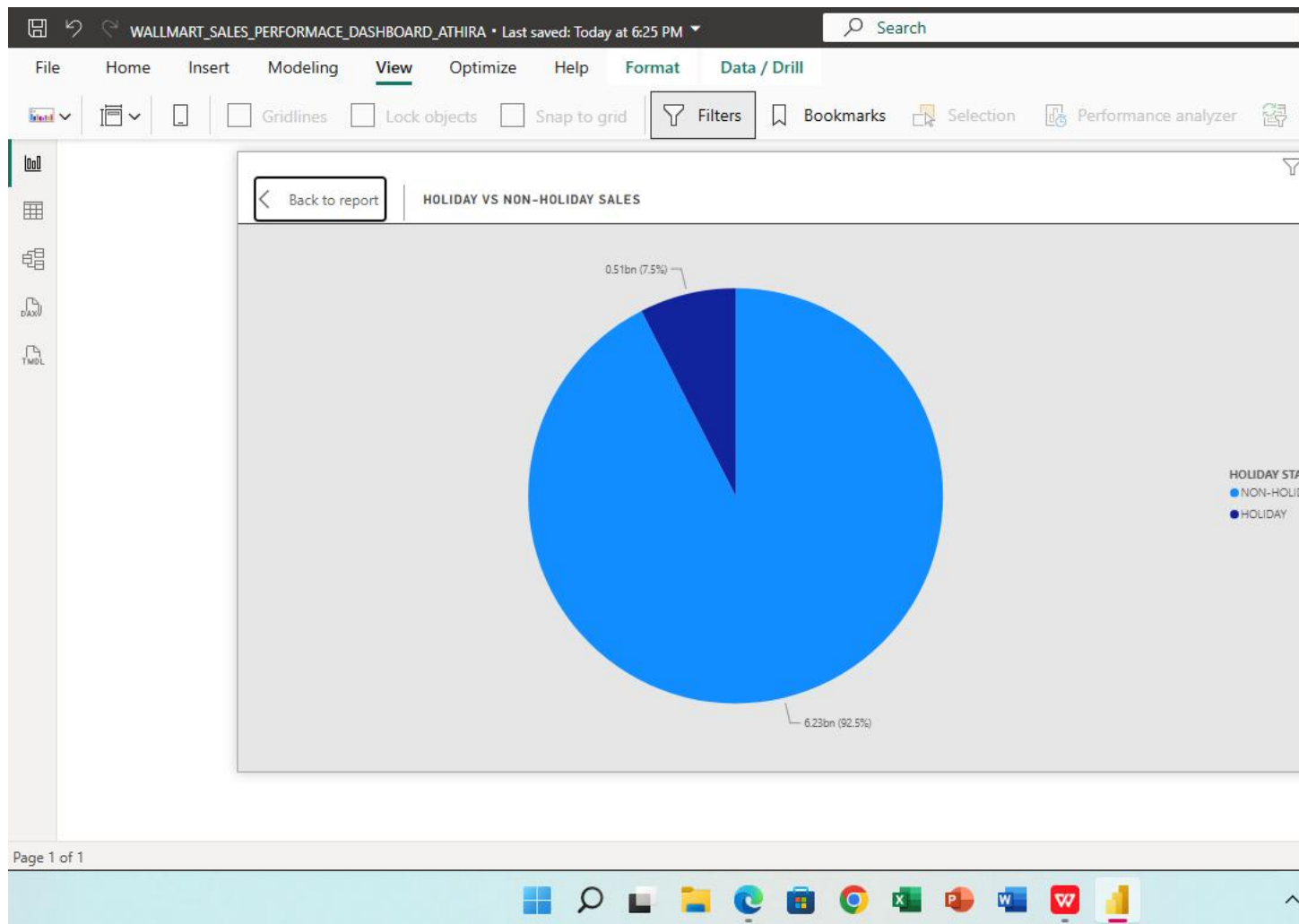
Sales by Store

A clustered bar chart lines up all 45 stores by total weekly sales. This was useful for spotting which stores were consistently outperforming the rest and which ones were lagging behind.



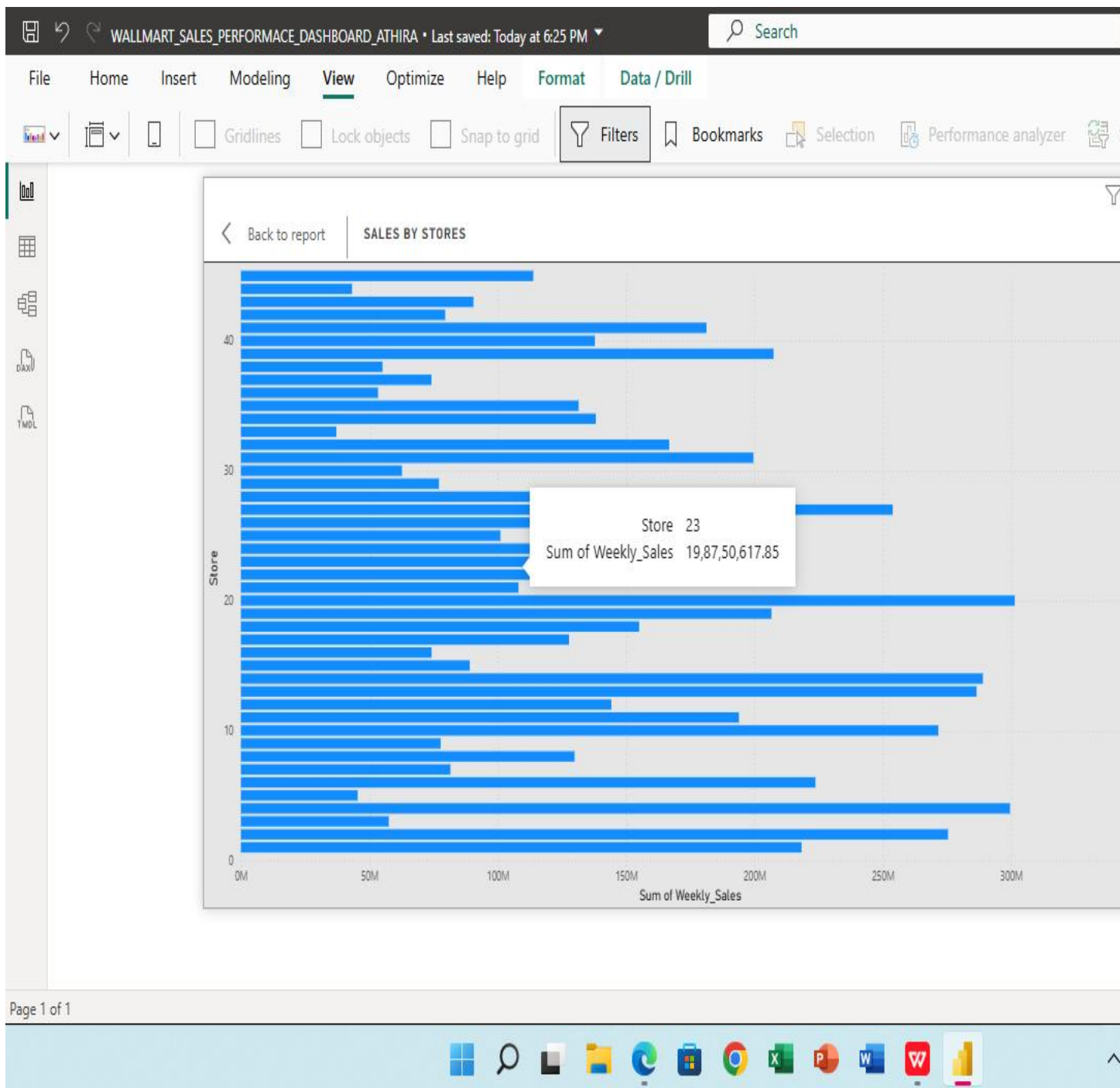
Holiday vs Non-Holiday Sales

A pie chart splits total sales between holiday and non-holiday weeks. Even though holiday weeks are a small fraction of the calendar, this chart still shows how much of a spike they cause.



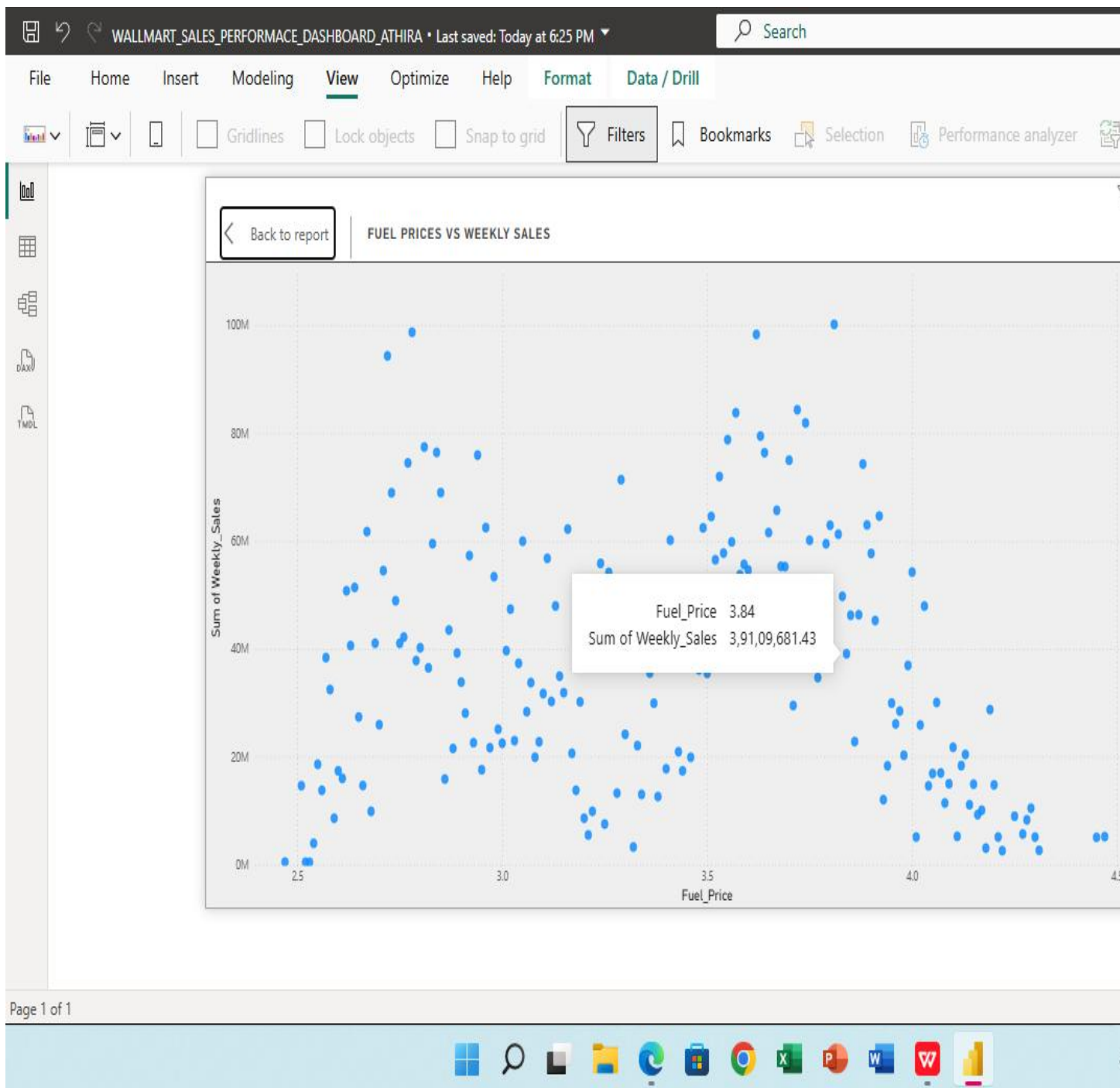
Sales Contribution by Store

I added a treemap to show each store's share of total sales — the bigger the tile, the bigger that store's contribution. I found this easier to read than the bar chart when I wanted a sense of proportion rather than exact rankings.



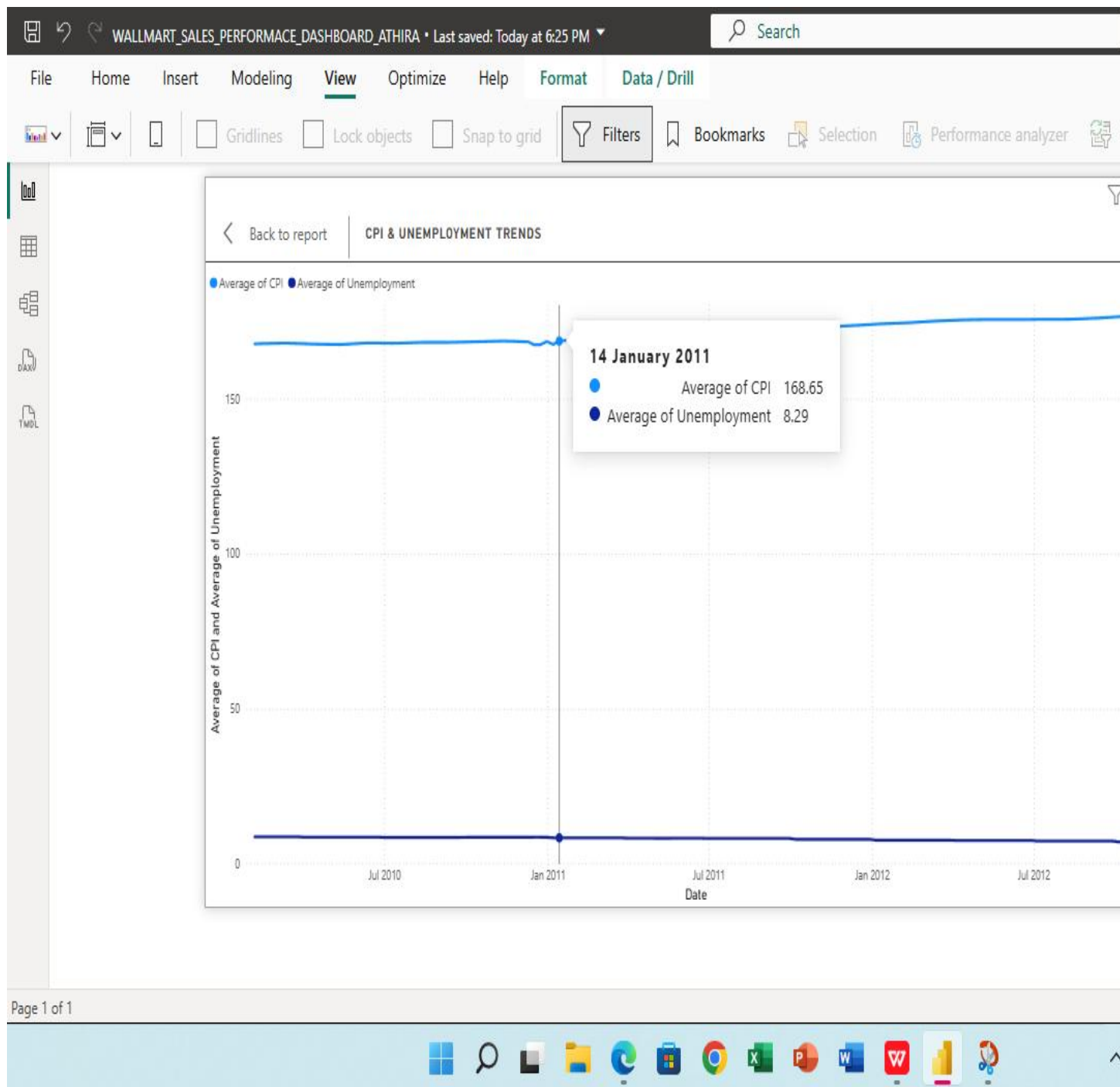
Fuel Price vs Weekly Sales

A scatter chart plots fuel price against weekly sales to see if there's any correlation between the two. The points come out fairly scattered with no clear pattern, which told me fuel price isn't a strong driver of sales in this dataset.



CPI and Unemployment Trends

A separate line chart tracks CPI and unemployment over time, sitting alongside the sales data so economic context isn't disconnected from the sales trends.



7. Slicers Used

To make the dashboard interactive rather than a fixed report, I added three slicers:

- Year
- Store
- Holiday Status

These let a user narrow the entire dashboard down to a specific year, a single store, or just holiday weeks, and every visual updates together.

8. Key Findings

A few things stood out once the dashboard was complete:

- Total sales across all stores come to roughly \$6.73 billion.
- Average weekly sales sit around \$1.05 million.
- The dataset covers 45 stores in total.
- The single highest weekly sales figure recorded was about \$3.82 million.
- Sales performance varies quite a bit from store to store.
- Non-holiday weeks bring in more total sales than holiday weeks, simply because there are far more of them.
- Weekly sales fluctuate noticeably across the year rather than staying flat.
- Fuel price doesn't show any strong relationship with weekly sales in this dataset.

9. Conclusion

Working on this project gave me a much better feel for how Power BI turns a plain spreadsheet into something people can actually use to make decisions. Picking the right chart for each question mattered more than I expected going in — the treemap and pie chart told a very different story from the line chart, even though they came from the same underlying numbers.

Adding the slicers was probably the most satisfying part, since it changed the dashboard from something you just look at into something you can actually explore. Overall, this project helped me understand not just Power BI as a tool, but how a well-built dashboard can make Walmart's sales performance genuinely easier to read and act on.

10. References

- Microsoft Power BI Desktop
- Walmart Sales Dataset
- Microsoft Power BI Documentation